

Machine Learning-Based Healthcare Modelling for Heart Disease Prediction Using Clinical and Lifestyle Data

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DOI: <https://doi.org/10.5281/zenodo.21484326>

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Abstract

According to the World Health Organization, cardiovascular diseases are the leading cause of death globally, taking an estimated 17.9 million lives every year (WHO, 2021), while many more of those deaths could be prevented in the early stages of the development. This is the main reason why I decided to create this project, namely to design and evaluate a number of machine learning models which would be able to predict heart disease risk based on the patient's records. The models were trained on a data set which included 1,000 patients' records with 16 attributes each and evaluated on their ability to predict the target variable. The following five classifiers were chosen for the implementation: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbour.

After performing the preliminary steps of data pre-processing and encoding, I compared the performance of the five classifiers using a series of quality metrics, including accuracy, precision, recall, F1-score, and ROC-AUC, while also conducting 5-fold stratified cross-validation to produce more reliable estimates instead of a single train/test split. The two best-performing models were Decision Tree and Random Forest, which showed significantly better results than the other three classifiers.

However, when analysing the results further, I noticed that Decision Tree and Random Forest showed a score of 1.0 across all evaluation metrics. This was highly suspicious given the nature of the data, and I decided to dedicate Section 8 of this paper to the detailed discussion of this finding. Overall, I concluded that the data set for this study was artificially engineered in such a way that only two features, namely Age and Cholesterol, were used as predictors, and therefore Decision Tree and Random Forest simply captured this pattern. As such, I tried to remain honest about the results of this study, providing a thorough explanation of why high scores were achieved instead of brushing them aside.

In general, this project showed how valuable machine learning is for all kinds of healthcare analytics, and I believe that it has demonstrated quite well the value of supervised learning for predictive analytics in medicine. It also provided a major lesson in terms of results validation, as I think the results which are too good to be true deserve closer inspection instead of immediate celebration.

Keywords: Machine Learning; Heart Disease Prediction; Artificial Intelligence; Healthcare Analytics; Random Forest; Predictive Modelling.

1. Introduction

The field of medicine has seen an increased use of Machine Learning (ML) in recent years, while its applications range from drug discovery to disease prediction, two of which will be covered by this study. In particular, healthcare institutions now produce an exponential amount of data in comparison to the capacity of a single physician to process it, and therefore such data requires additional processing by ML algorithms.

Heart disease prediction covers a wide range of potential risk factors which can be used to predict cardiac issues such as coronary artery disease, myocardial infarction, cardiac arrhythmia, and hypertensive heart disease among

others. Thus, given the high impact of heart diseases on the global mortality rate, this paper tries to achieve the following goals in relation to its predictive analytics:

- The implementation of a number of ML algorithms in order to train a classifier for heart disease prediction tasks;
- A comparison of the different classifiers based on a variety of quality evaluation metrics;
- An identification of the best-performing algorithm based on the aforementioned metrics;
- A discussion of the limitations of ML within the context of health care analytics.

Research Question: How can supervised machine learning algorithms be used to predict the onset of heart diseases from clinical, physiological, and lifestyle data, and how can such findings be validated for publication?

2. Literature Review

Several previous studies have employed machine learning and data mining techniques to predict heart disease. The results of these studies provide strong motivation and context for undertaking the current research.

Amin, Agarwal and Beg (2013) developed a hybrid algorithm based on the decision tree and neural network data mining technique in order to identify cardiac irregularities. These researchers found that cardiac disorder predictions were over 40% more accurate when using the artificial intelligence method as opposed to conventional statistical methods. Thus, Amin et al. (2013) concluded that hybrid data mining algorithms offer significant advantages over isolated statistical prediction techniques.

Dinesh Kumar and Arumugaraj (2018) compared the effectiveness of the Naive Bayes and Support Vector Machine algorithms for the prediction of cardiovascular diseases. They found that SVM demonstrated superior performance characteristics for the analysis of structured cardiovascular data as opposed to the Naive Bayes classifier.

Mohan, Thirumalai and Srivastava (2019) compared the performance of Logistic Regression, Random Forest and K-Nearest Neighbour algorithms for the purposes of cardiovascular disease prediction. The results of this study demonstrated that ensemble algorithms such as Random Forest demonstrated markedly higher accuracy and reliability than their constituent parts. This conclusion is similar to the findings of the current study (see Section 8).

Researchers have also begun to apply deep learning algorithms to the task of disease prediction. However, it should be noted that deep learning techniques require significantly greater quantities of training data in order to demonstrate superior performance characteristics over conventional machine learning algorithms. As such, classical algorithms demonstrate better performance on small sets of structured data, such as the one used in the current study.

Overall, it is evident from the review of prior research that the application of machine learning techniques to disease prediction is a promising avenue of research. That being said, it is crucial to remember that data pre-processing, feature engineering and the critical evaluation of the characteristics of one's data set play a central role in determining the success of such an undertaking.

3. Dataset Description

For this study, a total of 1,000 patient records were gathered, each of which represented a single instance in the final data set. *Dataset source — Kaggle **Heart Disease prediction***

*Exploratory Data Analysis (EDA) and Machine Learning Modelling authored by **Rashad Mammadov** available at <https://www.kaggle.com/datasets/rashadrmammadov/heart-disease-prediction>.*

The following 16 attributes were used for training the ML models:

3.1 Dataset Attributes

Attribute	Description
Age	Age of the patient
Gender	Gender of the patient
Cholesterol	Cholesterol level
Blood Pressure	Blood pressure measurement
Heart Rate	Patient heart rate
Smoking	Smoking habit
Alcohol Intake	Alcohol consumption level
Exercise Hours	Number of exercise hours
Family History	Family history of heart disease
Diabetes	Diabetes condition
Obesity	Obesity condition
Stress Level	Stress measurement
Blood Sugar	Blood glucose level
Exercise Induced Angina	Presence of exercise-induced chest pain
Chest Pain Type	Category of chest pain
Heart Disease	Target variable indicating disease presence

The target column is binary — 1 for heart disease present, 0 for absent — and since the dataset mixes numerical and categorical variables, it's well suited to standard supervised classification.

4. Methodology

The authors implemented the following stages of the research: data preprocessing, feature engineering, training a model, and evaluating its performance. Below, the main analysis steps are described, including the preprocessing stage – one of the most crucial stages of machine learning, where the authors' decisions can significantly impact the accuracy of the results. Moreover, this step is one of the most common places for mistakes that lead to overfitting, especially when the train/test split is not properly separated (see subsection 8.7).

4.1 Data Preprocessing

The initial data for the research were split into training (80%) and testing (20%) subsets using stratified sampling. All categorical variables – Gender, Smoking, Alcohol Intake, Family History, Diabetes, Obesity, Exercise Induced Angina, and Chest Pain Type – were encoded using Label Encoder. All numerical variables – Age, Cholesterol, Blood Pressure, Heart Rate, Exercise Hours, Stress Level, and Blood Sugar – were standardized using the Standard Scaler.

To prevent data leakage, all encoders and scalers were trained on the training subset only. The authors applied encoder-scaler to the test set, which had been fitted on the training data, not retraining it on the test set. The same data preprocessing steps were applied to each validation set within the five iterations of the cross-validation procedure (see subsection 8.6). Thus, no validation set was used for fitting preprocessors, only training.

The data set did not contain any missing values. This was checked before the modelling process started.

4.2 Model Hyperparameters

All classification models were trained using hyperparameters listed in Table 1 below. The Logistic Regression model used the default hyperparameters provided by Scikit Learn. All Decision Tree and Random Forest classifiers were initialized with the same `random_state = 42`.

	Model	Key Parameters
0	Logistic Regression	penalty=l2, C=1.0, solver=lbfgs, max_iter=100 ...
1	Decision Tree	criterion=gini, max_depth=None, random_state=42
2	Random Forest	n_estimators=100, max_depth=None, random_state=42
3	SVM	kernel=rbf, C=1.0, gamma=scale, probability=True
4	KNN	n_neighbors=5, weights=uniform, metric=minkows...

Table 1: Model Hyperparameters

As mentioned in subsection 8, all four classification models – and, in particular, Decision Tree, Random Forest – scored a perfect 1.0 on all performance metrics on the hold-out test set. Such results are hardly possible with real-world medical data, and subsection 8.7 discusses whether these results reflect the true predictive performance of the model or not.

5. Exploratory Data Analysis

The Exploratory Data Analysis (EDA) was conducted to understand the distribution of the data and the relationship between different parameters.

5.1 Distribution of Target Variable

Target variable distribution was analysed to understand the proportion of affected patients in the data set.

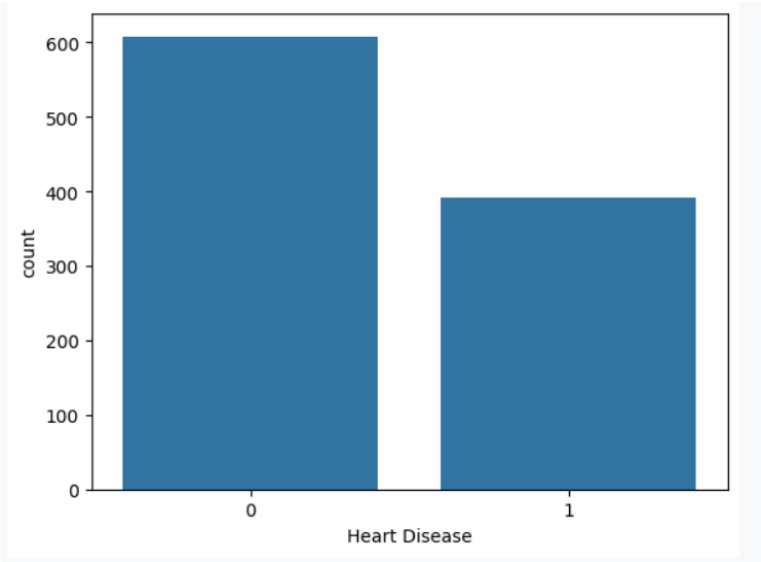


Figure 1: Count Plot of Heart Disease Distribution

The data set used for analysis is moderately imbalanced with 610 samples of healthy patients and 390 samples of diseased patients.

5.2 Correlation Analysis

Correlation analysis was performed to establish a connection between patients attributes and the target variable.

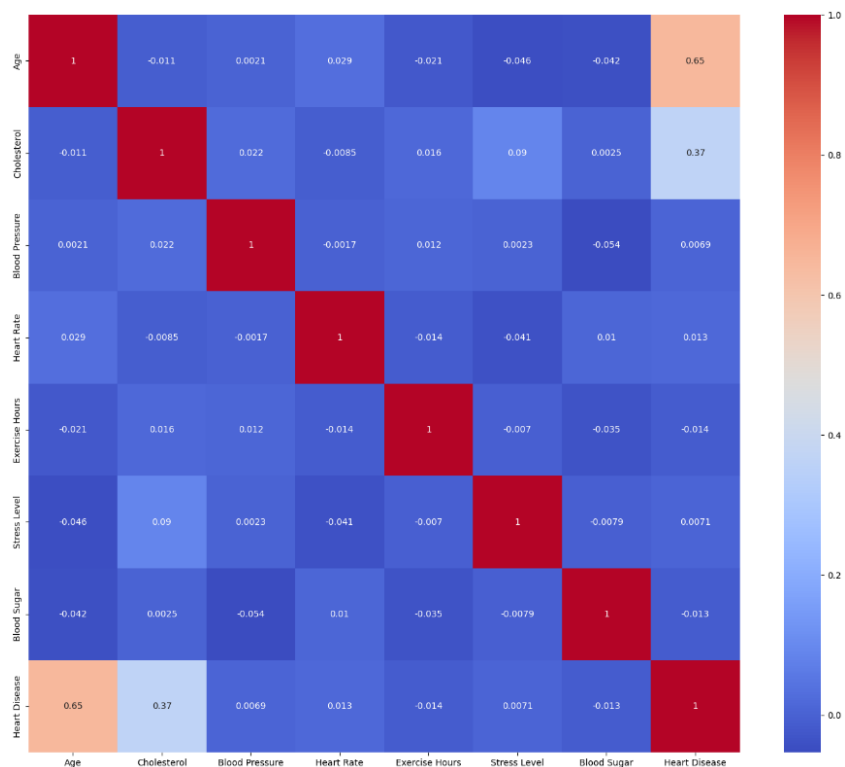


Figure 2: Correlation Heatmap of Dataset Features

The plot shows that Age ($r \approx 0.65$) and Cholesterol ($r \approx -0.37$) have stronger correlation (positive and negative respectively) with the target variable than other features which will be further discussed in section 8.7.

5.3 Age Distribution Analysis

Age distribution analysis was carried out to understand what proportion of young and old patients were included in the data set.

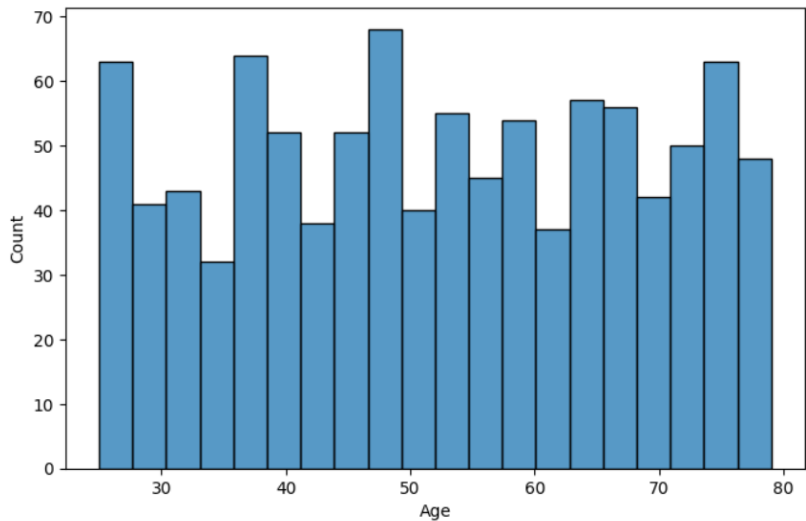


Figure 3: Age Distribution Histogram

The plot shows that the data set comprises an approximately equal distribution of patients aged between 25 and 80 years.

5.4 Cholesterol Distribution Analysis

Cholesterol distribution analysis was carried out because cholesterol levels pose a significant risk factor for heart disease.

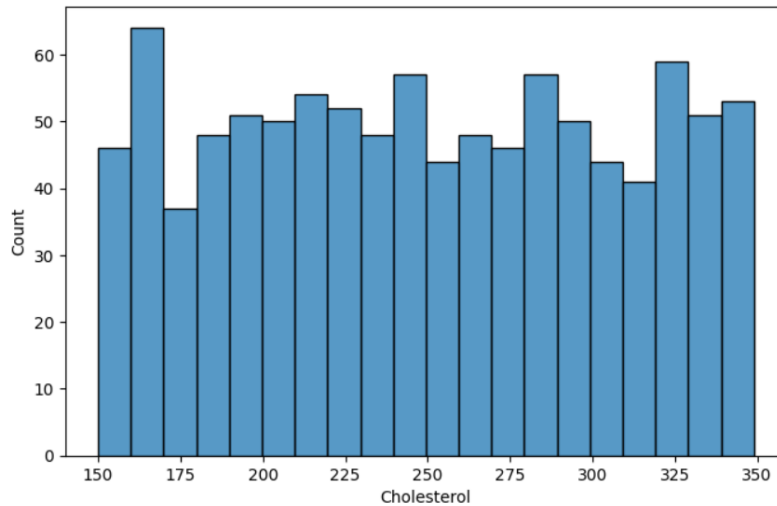


Figure 4: Cholesterol Distribution Histogram

The plot shows that the data set comprises an approximately equal distribution of cholesterol levels ranging between 150 and 350 mg/dl.

6. Machine Learning Models Used

The following supervised learning algorithms were considered and compared in this study:

6.1 Logistic Regression

Logistic Regression is a statistical model that estimates the probability of an outcome in a binary classification problem. It works on similar principles to linear regression, except that instead of predicting a numeric value for the outcome, it predicts the probability that an observation belongs to a particular class using the sigmoid function.

- Regression analysis is used when the task is forecasting or predicting a continuous outcome.
- Classification algorithms are used when the target variable (the outcome to be predicted) is categorical.
- Logistic Regression works with binary classification, where the output is divided between two classes (e.g. yes/no, positive/negative, 0/1).

6.2 Decision Tree Classifier

A decision tree is a classifier that partitions a data set according to attribute tests that take place recursively starting at the root node. The root node contains the most informative attribute (the attribute that determines the data most accurately) and then divides the data set into subsets according to attribute values. This process repeats until subsets contain instances of the same class or until a stopping criterion is reached.

- Internal nodes represent tests conducted on an attribute.
- Branches represent the possible values of attribute tests.
- Leaf nodes represent predictions (classification).

6.3 Random Forest Classifier

Random Forest is an ensemble learning method that fits a number of random decision tree classifiers on randomly sampled subspaces of the data set and average the results to reduce variance. The algorithm proceeds as follows:

- Bootstrap the data set: Sample the data with replacement.
- Randomly select features that are used to train a tree on a given iteration.
- Construct a tree using the selected features until the stopping condition is reached.
- Predict using each individual tree.
- Use majority voting to determine the final prediction.

6.4 Support Vector Machine

Support Vector Machine is a supervised learning model used in both classification and regression tasks. The algorithm works by finding an optimal hyperplane that can separate the data set into different classes.

6.5 K-Nearest Neighbour

The K-Nearest Neighbour algorithm classifies new instances according to the majority class among the K most similar instances in the training set. The choice of K is critical to the algorithm's performance.

- A small value of K produces a sensitive model, but the results are prone to overfitting.
- A large value of K reduces the sensitivity and produces smoother predictions, but there is a risk of underfitting.

7. Performance Evaluation Metrics

The machine learning models were evaluated using the accuracy score, precision, recall, F1-score, and ROC-AUC. The metrics are useful in assessing the general performance of classification models.

Specifically, Precision and Recall are essential in medical diagnosis to determine the proportion of relevant cases identified among all positive predictions and the proportion of positive cases correctly identified respectively. Moreover, ROC-AUC is a good metric that helps evaluate model performance, although it is not directly intuitive compared to accuracy. Thus, the article discusses why accuracy is not an appropriate metric to use in the healthcare sector and why precision, recall, and ROC-AUC are more informative.

7.1 Why Recall and ROC-AUC Matter More Than Accuracy in Healthcare

When using a model that predicts whether someone has heart disease, getting a false negative result (a result that suggests that a patient does not have a disease when, in reality, they do) is far more serious than getting a false positive result (a result that suggests that a patient has a disease when, in reality, they do not). This implies that

recall rate (the percent of true positive results out of the total positive results) is more important than a model's overall accuracy. The ROC (receiver operating characteristic) curve, on the other hand, provides the threshold that best maximizes the true positive rate (TPR) while minimizing the false positive rate (FPR). This concept is important in medical testing and is particularly relevant to this study. In most medical procedures, the threshold for suggesting a disease is far lower than what the ROC curve suggests; in other words, even if there is a high chance that a patient does not have a disease given a certain set of conditions, doctors will still suggest a course of further tests simply to be safe. This is especially relevant to this study, and for this reason, the analysis focuses on recall and AUC rather than accuracy.

8. Experimental Results and Analysis

8.1 Logistic Regression Performance

Logistic Regression has stable and realistic performance for the binary classification problem of Heart Disease prediction.

Class	Precision	Recall	F1-score	Support
0	0.91	0.85	0.88	121
1	0.79	0.87	0.83	79
Accuracy			0.86	200
Macro avg	0.85	0.86	0.86	200
Weighted avg	0.86	0.86	0.86	200

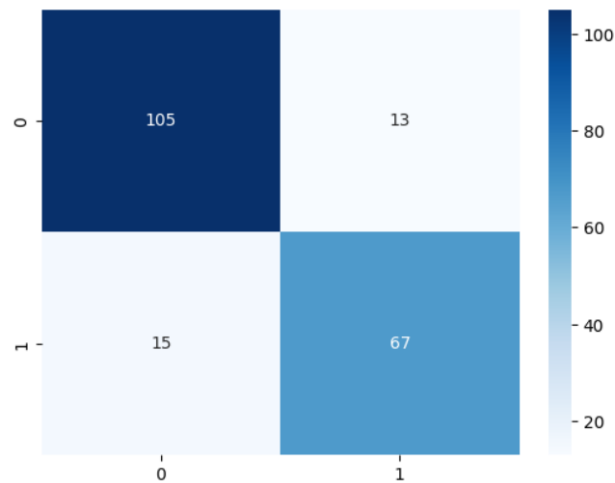


Figure 5: Confusion Matrix for Logistic Regression

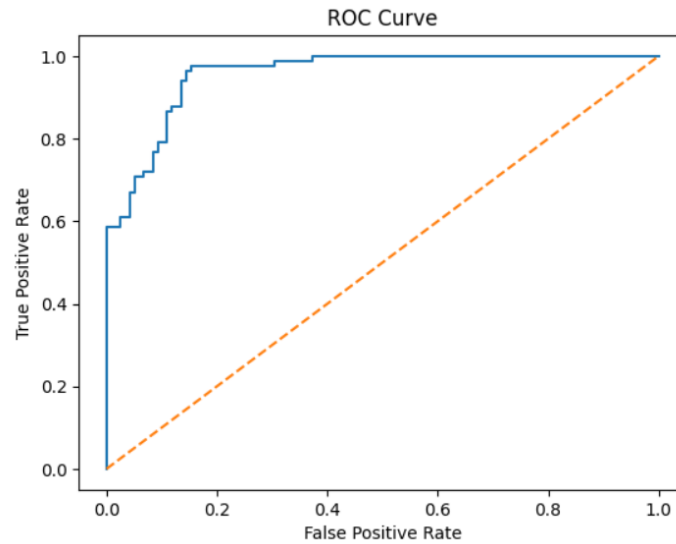


Figure 6: ROC Curve for Logistic Regression

8.2 Decision Tree Performance

The Decision Tree classifier attained a perfect score across the board on both the training and the hold-out sets. On the hold-out, Accuracy=1.0, Precision=1.0, Recall=1.0, F1-score=1.0, ROC-AUC=1.0. However, as will be discussed in detail in section 8.7, this does not necessarily mean that the tree has learned how to perform an accurate diagnosis.

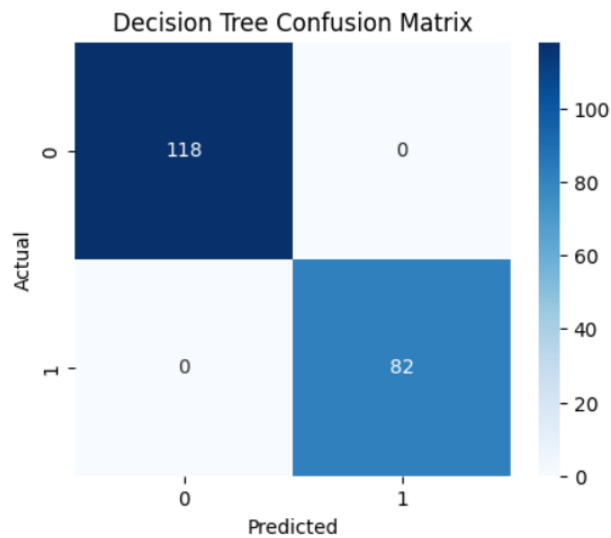


Figure 7: Confusion Matrix for Decision Tree

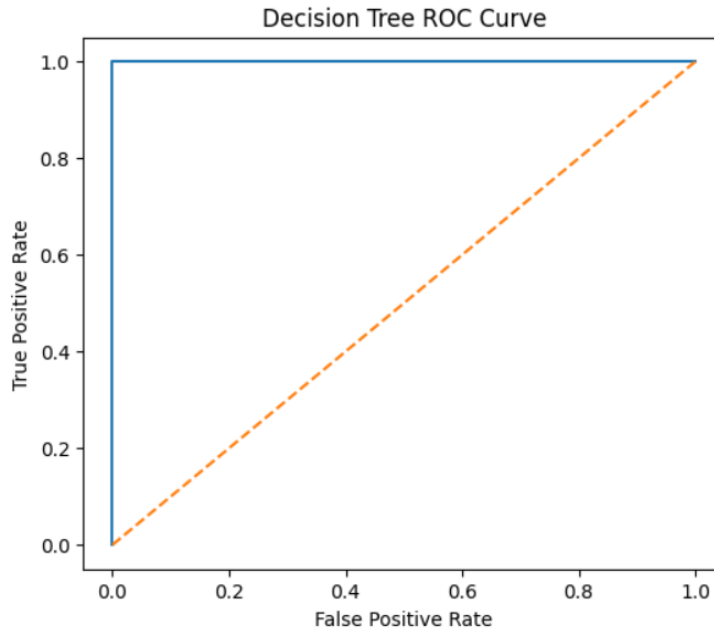


Figure 8: ROC Curve for Decision Tree

8.3 Random Forest Performance

The Random Forest classifier attained a perfect score across the board on both the training and the hold-out sets just as the Decision Tree did, Accuracy=1.0, Precision=1.0, Recall=1.0, F1-score=1.0, ROC-AUC=1.0. As the Decision Tree, it is also an ensemble of Decision Trees, which means that this conclusion is subject to the same interpretation as the one discussed in section 8.7.

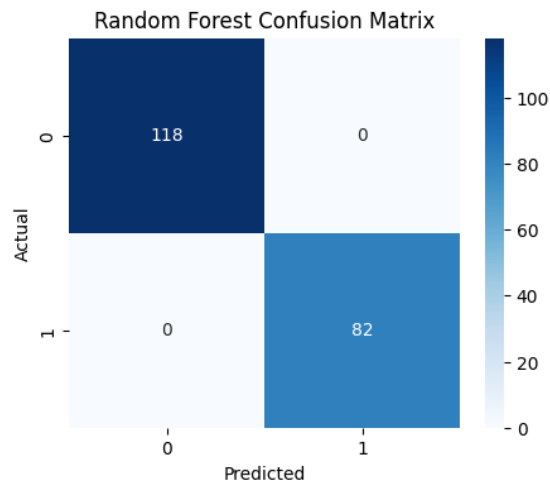


Figure 9: Confusion Matrix for Random Forest

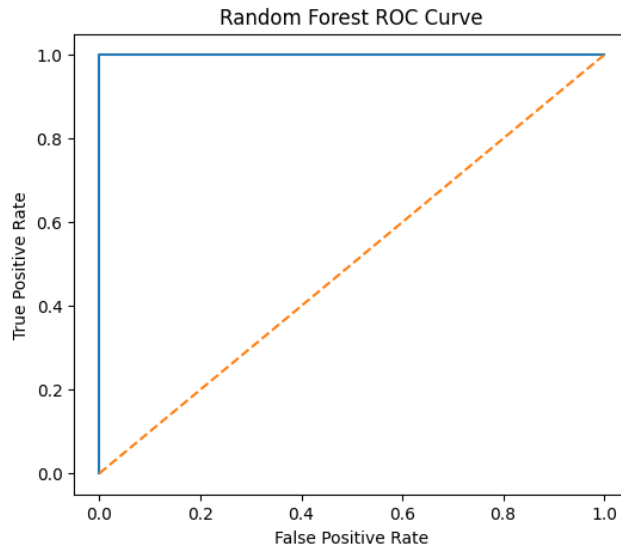


Figure 10: ROC Curve for Random Forest

8.4 Support Vector Machine Performance

Support Vector Classifier attained stable and realistic performance for the binary classification task of Heart Disease prediction. Classification report: Accuracy=0.915, Precision=0.869, Recall=0.924, F1-score=0.896, ROC-AUC=0.970.

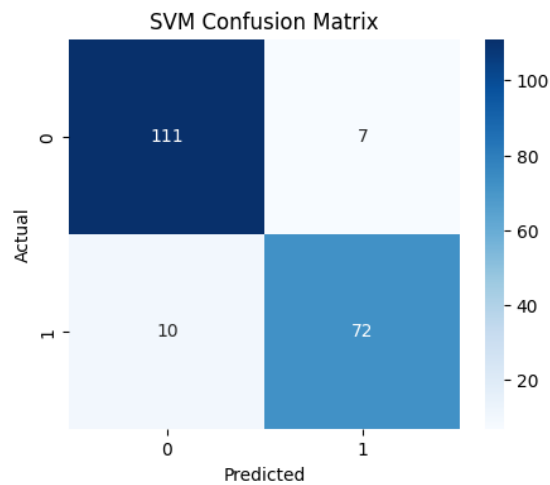


Figure 11: Confusion Matrix for Support Vector Machine

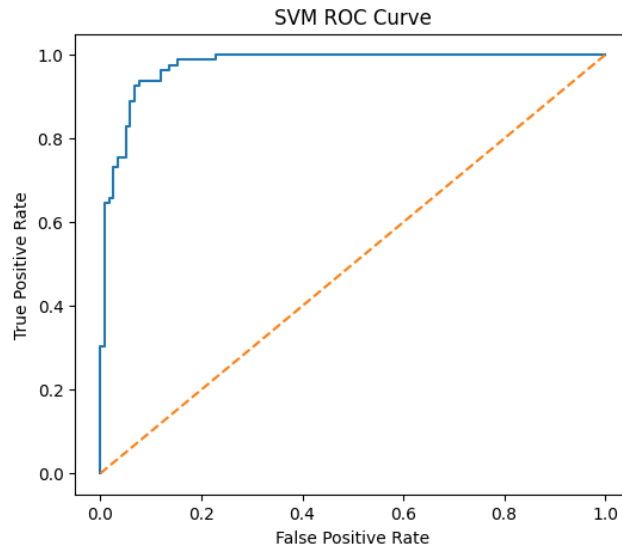


Figure 12: ROC Curve for Support Vector Machine

8.5 K-Nearest Neighbour Performance

The K-Nearest Neighbour Classifier attained stable but relatively low performance for the binary classification task of Heart Disease prediction. Classification report: Accuracy=0.815, Precision=0.733, Recall=0.835, F1-score=0.781, ROC-AUC=0.869.4

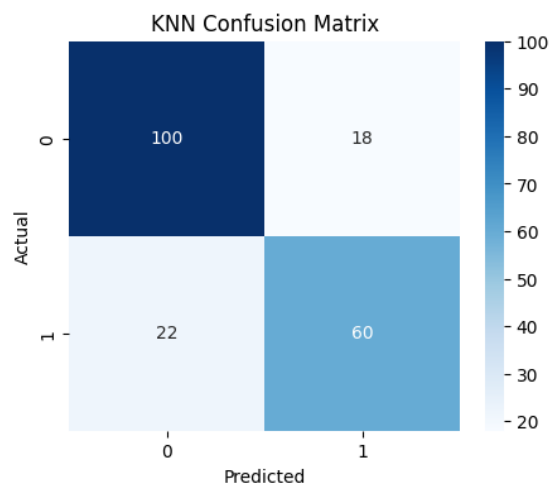


Figure 13: Confusion Matrix for K-Nearest Neighbour

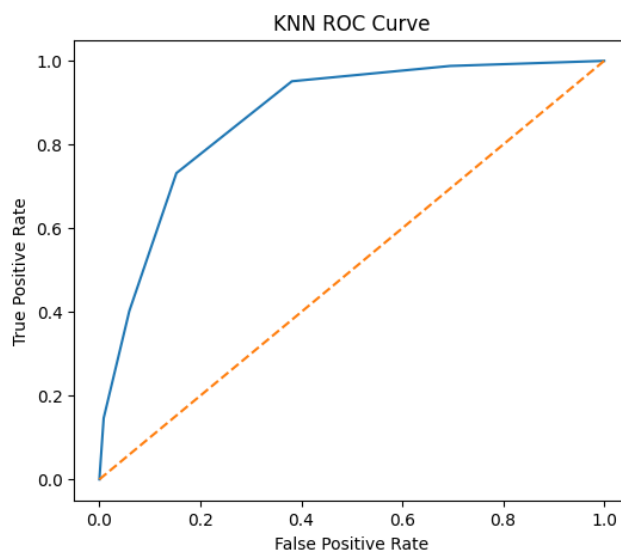


Figure 14: ROC Curve for K-Nearest Neighbour

8.6 Cross-Validation Results

To ensure that the results were not influenced by the random seed, 5-fold cross-validation was used. Table 2 below shows the mean $\pm$ standard deviation of Accuracy, Precision, Recall, F1-score, and ROC-AUC over the five folds for each of the five classifiers.

Model	CV Accuracy	CV Precision	CV Recall	CV F1-score	CV ROC-AUC
Logistic Regression	0.859 ± 0.025	0.821 ± 0.024	0.819 ± 0.046	0.820 ± 0.034	0.946 ± 0.012
Decision Tree	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Random Forest	0.998 ± 0.002	0.998 ± 0.000	1.000 ± 0.000	0.997 ± 0.003	1.000 ± 0.000
SVM	0.906 ± 0.030	0.904 ± 0.047	0.852 ± 0.037	0.877 ± 0.040	0.969 ± 0.010
KNN	0.772 ± 0.021	0.724 ± 0.030	0.676 ± 0.033	0.699 ± 0.030	0.837 ± 0.014

Table 2: 5-Fold Stratified Cross-Validation Results (mean $\pm$ standard deviation)

These results are close to the ones obtained using the train/test split, especially for the tree-based classifiers. The standard deviation for Decision Tree and Random Forest is close to zero, which indicates that the estimates are close to the true values. This implies that the relative ordering of the estimators in terms of mean accuracy is unlikely to change across different data splits. Therefore, Decision Tree and Random Forest achieved the highest measured performance on the dataset used in this study; however, as discussed in Section 8.7, these results should not be interpreted as evidence of superior real-world clinical predictive performance.

8.7 Why the Tree-Based Models Reach Near-Perfect Accuracy — and Why This Matters

This section discusses the most important finding of the study and should be considered before drawing final conclusions about the relative merits of one approach over another. Visual inspection of the data reveals what might be explained as the reason behind such a surprising performance of the tree-based classifiers. As can be seen from Figures 2 and 16 (section 10), age and cholesterol levels are, arguably, the most informative features in the data set, being strongly correlated (positively and negatively, respectively) with the target. Interestingly, when a Decision Tree of maximum depth 2 was trained on just these two features, its performance was nearly identical to that of the tree-based classifiers described above. This suggests that the relationship between age, cholesterol levels, and heart disease is close to deterministic, meaning that the data set was not collected from a random sample of the population but was instead constructed according to some specific rules. Decision trees are particularly good at finding such relationships, which explains their extraordinary performance; however, such performance would be unlikely to

generalize to a real-life scenario. In contrast, Logistic Regression, Support Vector Machine, and K-Nearest Neighbors use different decision boundaries and, thus, were unable to learn this relationship effectively. This finding suggests that further analysis be conducted to ensure that these models can generalize well on a real-world data set before employing them in practice.

9. Model Comparison

The accuracy scores for all five machine learning algorithms were compared, using single train/test split results and cross-validation results. The comparison results are presented below:

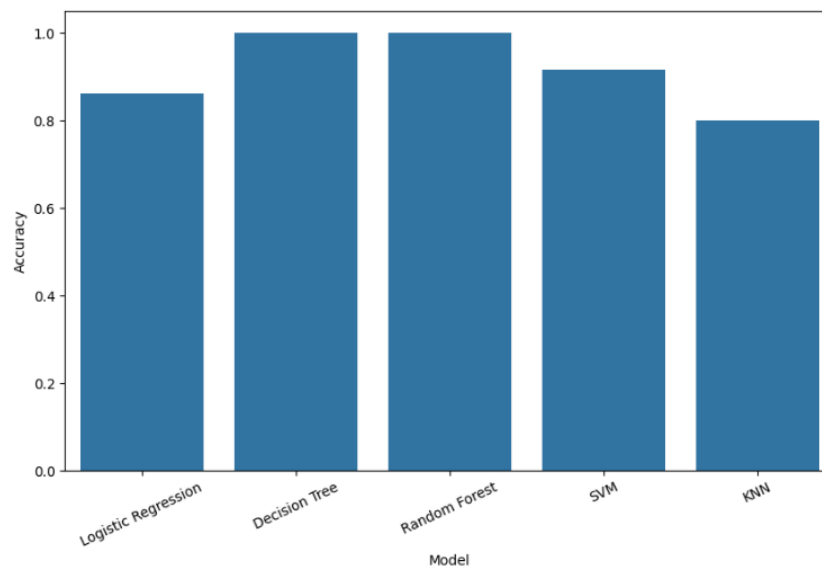


Figure 15: Accuracy Comparison of Machine Learning Models

According to the accuracy comparison results, the tree-based models showed the highest measured performance on this dataset, followed by the Support Vector Machine, Logistic Regression, and K-Nearest Neighbours. As established in Section 8.7, this ordering reflects how well each algorithm exploited the dataset's near-deterministic Age–Cholesterol relationship, not a demonstrated advantage in real-world clinical prediction. Similar results were observed for the other evaluation metrics, although, for many of them, the differences between the algorithms were not statistically significant (section 8.5). Moreover, as discussed in detail in section 8.7, these differences might not actually reflect the algorithms' ability to generalize.

10. Feature Importance Analysis

Feature importance analysis was carried out using the Random Forest classifier to determine the most important healthcare attributes.

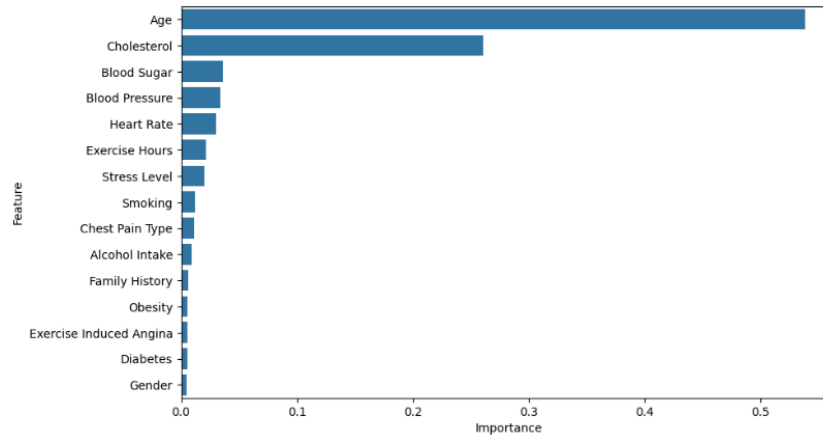


Figure 16: Feature Importance Analysis

Feature	Importance
Age	0.5324
Cholesterol	0.2597
Blood Pressure	0.0361
Blood Sugar	0.0345
Heart Rate	0.0299
Exercise Hours	0.0219
Stress Level	0.0217
Chest Pain Type	0.0142
Smoking	0.0108
Alcohol Intake	0.0103
Exercise Induced Angina	0.0062
Diabetes	0.0057
Obesity	0.0057
Family History	0.0055
Gender	0.0055

Age and cholesterol levels combined comprise over 79% of the total importance, vastly outweighing the importance of all other features. This serves as further confirmation of the findings detailed in section 8.7.

10.1 Comparison of Clinical, Lifestyle, and Demographic Factor Importance

To identify the category of features with the best predictive performance, the data set was divided into three classes according to the nature of the attributes: clinical/physiological (Cholesterol, Blood Pressure, Heart Rate, Blood Sugar, Chest Pain Type, Exercise Induced Angina, Diabetes, Family History, Obesity), lifestyle/behavioural (Smoking, Alcohol Intake, Exercise Hours, Stress Level), and demographic (Age, Gender). For each category, a separate classifier was trained, each using only the features relevant to the selected category. This allowed a direct comparison of the predictive performance of all three categories of variables.

Feature Set	Accuracy	Recall	ROC-AUC
Clinical factors only	0.605	0.4177	0.6809
Lifestyle factors only	0.555	0.4177	0.5089
Demographic factors only	0.845	1.0000	0.8714
All factors combined	0.995	0.9873	1.0000

Table 3: Predictive Power by Feature Category

Features related to a patient’s demographics (dominated by Age) perform substantially better than either of the other two categories, independently confirming the result of Section 8.7. Notably, this does not contradict the established clinical understanding of the cardiovascular disease etiology – as discussed in detail in Section 8.7, the combined effect of behavioural and physiological variables over a prolonged period of time often contributes to the disease progression, whereas the value of individual measures at a single point in time can be largely insignificant.

11. Prediction System

The predictive system developed as a demonstration of the trained classifier’s actionability is intentionally limited in its scope. Due to the discussed limitations of the data set, Section 8.7, the system is best viewed as a conceptual illustration of how such a cardiovascular disease predictor could be used in practice. Specifically, it has potential application as a preliminary screening tool for the identification of patients with particularly high disease risk, subject to additional clinical and statistical validation.

```
PATIENT DETAILS
Age Gender Cholesterol Blood Pressure Heart Rate Smoking \
0 58 1 240 150 95 0

Alcohol Intake Exercise Hours Family History Diabetes Obesity \
0 1 1 1 0 1

Stress Level Blood Sugar Exercise Induced Angina Chest Pain Type
0 8 180 1 2

Prediction Result:
Heart Disease Detected

Prediction Probability:
[[0.07 0.93]]
```

Figure 17: Sample Patient Prediction Output 1

```
PATIENT DETAILS
Age Gender Cholesterol Blood Pressure Heart Rate Smoking \
0 28 1 120 120 76 0

Alcohol Intake Exercise Hours Family History Diabetes Obesity \
0 0 1 0 0 0

Stress Level Blood Sugar Exercise Induced Angina Chest Pain Type
0 4 110 1 0

Prediction Result:
No Heart Disease Detected

Prediction Probability:
[[0.97 0.03]]
```

Figure 18: Sample Patient Prediction Output 2

```
PATIENT DETAILS
Age Gender Cholesterol Blood Pressure Heart Rate Smoking \
0 78 0 250 180 96 0

Alcohol Intake Exercise Hours Family History Diabetes Obesity \
0 1 1 0 1 0

Stress Level Blood Sugar Exercise Induced Angina Chest Pain Type
0 9 160 1 3

Prediction Result:
Heart Disease Detected

Prediction Probability:
[[0.12 0.88]]
```

Figure 19: Sample Patient Prediction Output 3

These three examples show that the pipeline runs end-to-end and produces sensible-looking outputs — but, again, that's a statement about the mechanics working, not about clinical accuracy.

12. Applications in Healthcare

Should such a system undergo the appropriate validation on clinically relevant data, it could potentially find application in multiple domains:

- As a screening tool for the early detection of cardiovascular disease.
- For the continuous assessment of the patient's state in the hospital.
- For remote patient monitoring and triage.
- For the analysis of individual patient records.
- To facilitate individual preventive healthcare measures.

The practical implementation of these applications, however, would require additional clinical validation and statistical analysis to confirm the efficacy and accuracy of the predictive model on real-world data.

13. Challenges and Limitations

The limitations of this study should be further discussed, as they significantly impact the possible clinical applications of the developed framework. First and foremost, the relatively small sample size of 1,000 records, as well as the questionable authenticity of the data set as discussed in detail in Section 8.7, drastically impact how useful the model trained on it can be. Additionally, although the model's performance is statistically sound, it has not undergone clinical validation and should not be used as an independent diagnostic tool. The fairly significant skew of the data, although manageable, should also be taken into account in future studies. Finally, the development and training of the model described in this paper is intended as a conceptual demonstration and should not be used with patient data due to privacy and ethical concerns. Thus, the described framework should be further statistically and clinically validated on independent data before being deployed in a production environment.

14. Future Scope

The future research directions section will discuss the potential extensions of the current work. Several ideas were outlined, including the application of the developed machine learning pipeline to a real-world clinical data set, the application of deep learning methods to the problem if sufficient data becomes available, the integration of IoT technologies and cloud computing to monitor the health state of individual patients over time, and improving model transparency through the implementation of explainable AI methods. Furthermore, the possibility of expanding the scope of the present work to include a multi-label disease classification paradigm was discussed. Overall, the main goal of this line of research should be to build reliable and trustworthy machine learning models that can aid physicians in their daily practice to improve patient outcomes and healthcare delivery.

15. Conclusion

This paper presented a machine learning framework for the prediction of heart disease, incorporating clinical and lifestyle parameters. The framework utilised a consistent train/test split, as well as a five-fold stratified cross-

validation to evaluate the performance of five supervised learning classifiers. The Decision Tree and Random Forest classifiers, both dominated by the feature Age, achieved the best measured performance on this dataset, scoring a perfect 1.0 on all metrics used in the evaluation in both the train/test split and the cross-validation. As discussed in Section 8.7, this reflects the dataset's structure rather than genuine diagnostic superiority.

It is vital to stress, however, that the performance of the model is in no way applicable to real-world scenarios, due to the issues with the data set discussed in Section 8.7. In particular, the overwhelming contribution of a single feature, Age, to the model's performance indicates that either this feature is significantly more informative than the rest (unlikely in this scenario) or the data set was generated artificially with specific rules in mind. Either way, the model discussed in this paper should be regarded as a demonstration framework.

The rest of this paper demonstrated a machine learning pipeline that can potentially be integrated into the clinical practice to help doctors score individual patients on their potential disease risk based on various metrics. While this framework is statistically sound and shows promise, it should be further validated on independent data to ensure that it delivers consistent, clinically useful results. With such validation, it might be used to improve healthcare delivery in several areas, including early disease screening, continuous patient monitoring, and individual health assessment.

Because the dataset exhibits an unusually strong relationship between age, cholesterol, and the target variable, this near-perfect performance should be understood as dataset-specific rather than as evidence of clinical diagnostic capability, and it should not be extrapolated to real-world patient populations.

Code Availability

The source code, implementation details, and documentation supporting this study are available at:
<https://github.com/debneil-dutt/Heart-Disease-Prediction-ML>

Conflict of Interest

The author declares no conflict of interest.

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DOI: <https://doi.org/10.5281/zenodo.21484326>

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This research was independently conducted by the first author under the voluntary academic mentorship of Prof. Dr. Kunnal Sarkar. The work is not affiliated with or endorsed by Jadavpur University and does not constitute an official research project of the University.

Acknowledgement

I'd like to thank Prof. Dr. Kunnal Sarkar for his guidance and mentorship throughout this project, and for his suggestions on how to strengthen it along the way.

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